

Some hints or partial solutions for problem set 05 are given here.

I did not proofread carefully, so please use with care; watch out for typographical or other errors.

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1. The electrostatic field is $\mathbf{E} = \alpha x \hat{i} + \alpha y \hat{j}$ in some region of space. Here α is a positive constant.

(a) Show that this satisfies the requirement of zero curl.

(Partial) Solution/Hint $\rightarrow$

$$\begin{aligned}\nabla \times \mathbf{E} &= \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{i} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \hat{j} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{k} \\ &= \left(\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(\alpha y) \right) \hat{i} + \left(\frac{\partial}{\partial z}(\alpha x) - \frac{\partial}{\partial x}(0) \right) \hat{j} \\ &\quad + \left(\frac{\partial}{\partial x}(\alpha y) - \frac{\partial}{\partial y}(\alpha x) \right) \hat{k} \\ &= 0\end{aligned}$$

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- (b) Using Maxwell's first equation, find the electric charge density ρ in this region of space. How does ρ vary in space in this region?

(Partial) Solution/Hint $\rightarrow$

$$\frac{\rho}{\epsilon_0} = \nabla \cdot \mathbf{E} = \alpha + \alpha = 2\alpha \quad \implies \quad \rho = 2\epsilon_0 \alpha$$

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- (c) Calculate the difference of electric potential between the points $(1, 5, -1)$ and $(1, 5, +1)$.

(Partial) Solution/Hint →

The line joining the two points is in the z direction. There is no electric field in the z direction. The difference of electric potential is therefore zero.

For such problems, you are strongly urged to draw pictures which show you the directions involved. Sometimes drawing the situation from two different angles helps to clarify/visualize the 3D situation better!

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- (d) Calculate the difference of electric potential between the points $(0, 5, 0)$ and $(2, 5, 0)$.

(Partial) Solution/Hint →

The line joining the two points is in the x direction. The potential difference is

$$-\int \mathbf{E} \cdot d\mathbf{l} = -\int_{x=0}^{x=2} (\alpha x \hat{i} + \alpha y \hat{j}) \cdot dx \hat{i} = -\int_0^2 \alpha x dx = -2\alpha$$

Actually, the sign is not very meaningful because the problem just says “difference between”, not specifying whether $V_P - V_Q$ or $V_Q - V_P$ is required, where P or Q are the two points. So, 2α would be an equally valid answer.

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2. A charged sphere has radius a_0 and volume charge density $\rho = \rho_0(r/a_0)$, where r is the distance from the center of the sphere.

- (a) What is the total electric flux through any closed surface surrounding the sphere?

Hint: Use Gauss' law (Gauss' dielectric flux theorem). You will need to first calculate the total charge on the sphere.

(Partial) Solution/Hint →

Total charge:

$$Q = \int_0^{a_0} 4\pi r^2 \rho(r) dr = 4\pi \frac{\rho_0}{a_0} \int_0^{a_0} r^3 dr = \frac{4\pi \rho_0}{a_0} \frac{a_0^4}{4} = \pi \rho_0 a_0^3$$

The total electric flux according to Gauss' law is

$$\frac{\text{total enclosed charge}}{\epsilon_0} = \frac{\pi \rho_0 a_0^3}{\epsilon_0}$$

Note: For situations with spherical symmetry, you can use $d^3\mathbf{r} \equiv 4\pi r^2 dr$. Why? Where does this come from?

Since $\rho(\mathbf{r})$ depends only on the distance r to the center of the sphere (which we can take to be the origin), the integrand $\rho(\mathbf{r})$ is spherically symmetric. Hence we have used $\int \rho(\mathbf{r}) d^3\mathbf{r} = \int 4\pi r^2 \rho(r) dr$.

This is a basic technique in 3D integration, and you should have had to use it in earlier problem sets. so if you are not comfortable with this, please work on it, and/or discuss this with your tutor or fellow students, etc.

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- (b) Using Gauss' law, find the electric field outside the sphere.

(Partial) Solution/Hint →

As Gaussian surface, we can choose a spherical gaussian surface Σ of radius r concentric to the charged sphere, with $r > a_0$, i.e., the gaussian surface encloses the complete charged sphere. By symmetry, the electric field has the same magnitude at all points of

the surface and points radially outwards, hence is perpendicular to the surface (parallel to the surface-vector $d\mathbf{S}$) at all points of the surface. Therefore the flux is

$$\oint_{\Sigma} \mathbf{E} \cdot d\mathbf{S} = E \times \text{area} = E(4\pi r^2)$$

We have already found that the electric flux through any surface enclosing the sphere is $\frac{\pi\rho_0 a_0^3}{\epsilon_0}$. Thus

$$E(4\pi r^2) = \frac{\pi\rho_0 a_0^3}{\epsilon_0} \implies E = \frac{\rho_0 a_0^3}{4\epsilon_0} \frac{1}{r^2}$$

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- (c) Using Gauss' law, find the electric field inside the sphere.

(Partial) Solution/Hint $\rightarrow$

To find the electric field inside the sphere, we construct a spherical gaussian surface Σ_2 of radius r concentric to the charged sphere, but now $r < a_0$. This surface does not enclose the complete sphere, but only part of it. The charge enclosed within this surface is

$$q_{\text{encl.}} = 4\pi \frac{\rho_0}{a_0} \int_0^r r^3 dr = \frac{4\pi\rho_0}{a_0} \frac{r^4}{4} = \frac{\pi\rho_0 r^4}{a_0}$$

To find the electric flux through this surface, we need to use symmetry. By symmetry, the electric field has the same magnitude at all points of the surface and points radially outwards, hence is perpendicular to the surface (parallel to the surface-vector $d\mathbf{S}$) at all points of the surface. Therefore the flux is

$$\oint_{\Sigma} \mathbf{E} \cdot d\mathbf{S} = E \times \text{area} = E(4\pi r^2)$$

Thus Gauss's law (flux = $\frac{1}{\epsilon_0} \times$ enclosed charge) gives

$$E(4\pi r^2) = \frac{q_{\text{encl.}}}{\epsilon_0} = \frac{\pi\rho_0 r^4}{\epsilon_0 a_0} \implies E = \left(\frac{\rho_0}{4\epsilon_0 a_0} \right) r^2$$

- (d) Plot the magnitude of the electric field as a function of r . The plot should extend from $r = 0$ to $r = 2a_0$.

(Partial) Solution/Hint →

We have found

$$E = \begin{cases} \left(\frac{\rho_0}{4\epsilon_0 a_0} \right) r^2 & \text{for } r < a_0 \\ \left(\frac{\rho_0 a_0^3}{4\epsilon_0} \right) \frac{1}{r^2} & \text{for } r > a_0 \end{cases}$$

The curve should be $\propto r^2$ between $r = 0$ and $r = a_0$, and then should look like $\propto 1/r^2$ for $r > a_0$. The two curves meet at $r = a_0$, because if we use either formula at $r = a_0$ we obtain the value $(\rho_0 a_0)/(4\epsilon_0)$.

- (e) Calculate the electric potential as a function of r , both outside the sphere and inside the sphere. You can take $r = \infty$ as the reference, i.e., the potential at infinite distance can be taken to be zero.

(Partial) Solution/Hint →

For $r > a_0$, the electric field is the same as that created by a point charge at the center

$$E = \frac{Q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{r} = \left(\frac{\rho_0 a_0^3}{4\epsilon_0} \right) \frac{1}{r^2} \hat{r}$$

Hence we know without any calculation that the electric potential for $r > a_0$ (outside the sphere) is the same as that created by a point charge at the center:

$$V = \frac{Q}{4\pi\epsilon_0} \frac{1}{r} = \left(\frac{\rho_0 a_0^3}{4\epsilon_0} \right) \frac{1}{r}$$

We can also calculate this explicitly:

$$V(\mathbf{r}) = - \int_{\text{infinite}}^{|\mathbf{r}|=r} \mathbf{E} \cdot d\mathbf{r}$$

If we choose the path of integration to be a radial line from infinity to $\mathbf{r}$, then

$$\mathbf{E} \cdot d\mathbf{r} = \left(\frac{\rho_0 a_0^3}{4\epsilon_0} \right) \frac{1}{r^2} dr$$

so that

$$V(r) = - \int_{\infty}^r \left(\frac{\rho_0 a_0^3}{4\epsilon_0} \right) \frac{1}{r^2} dr = - \left(\frac{\rho_0 a_0^3}{4\epsilon_0} \right) \int_{\infty}^r \frac{dr}{r^2} = \left(\frac{\rho_0 a_0^3}{4\epsilon_0} \right) \frac{1}{r}$$

This works at points outside the sphere, all the way down to the surface of the sphere. At the surface the potential is

$$V(a_0) = \left(\frac{\rho_0 a_0^2}{4\epsilon_0} \right)$$

Now, how about within the sphere? We need to integrate along a line from $r = a_0$ to the point inside the sphere. Again we choose a radial line.

$$\begin{aligned} V(r) - V(a_0) &= - \int_{a_0}^r \mathbf{E} \cdot d\mathbf{r} = - \int_{a_0}^r \left(\frac{\rho_0}{4\epsilon_0 a_0} \right) r^2 dr \\ &= \left(\frac{\rho_0}{12\epsilon_0 a_0} \right) (a_0^3 - r^3) \end{aligned}$$

so that

$$\begin{aligned} V(r) - \left(\frac{\rho_0 a_0^2}{4\epsilon_0} \right) &= \left(\frac{\rho_0 a_0^2}{12\epsilon_0} \right) - \left(\frac{\rho_0}{12\epsilon_0 a_0} \right) r^3 \\ \implies V(r) &= \left(\frac{\rho_0 a_0^2}{3\epsilon_0} \right) - \left(\frac{\rho_0}{12\epsilon_0 a_0} \right) r^3 \end{aligned}$$

Probably a good exercise to plot the electric potential as a function of r . The plot could extend from $r = 0$ to $r = 3a_0$.

3. Recall that the current through a surface (e.g., through a cross-section of a current-carrying wire) is the surface integral of the current density.
- (a) A current I is uniformly distributed over the cross-section of a wire. The cross-section is circular, with radius R . Find the current density J , as a function of I and R .

(Partial) Solution/Hint $\rightarrow$

$$J = \frac{I}{\pi R^2}$$

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- (b) Now consider instead the situation that the current density is larger near the outer region of the wire cross-section than at the center: $J = kr$, where r is the radial distance from the center of the wire. Find the current I , as a function of k and R . You will probably need to do a two-dimensional integral in polar coordinates. (In two-dimensional polar coordinates, the area element is $dS = r dr d\theta$.)

(Partial) Solution/Hint $\rightarrow$

$$\begin{aligned} I &= \int d^2r J(r) \\ &= 2\pi \int_0^R dr \, r \, J(r) \quad \left\{ \begin{array}{l} \text{because } J(r) \text{ is cylin-} \\ \text{drically symmetric} \end{array} \right. \\ &= 2\pi \int_0^R dr \, r \, kr = 2\pi k \int_0^R dr \, r^2 = \frac{2\pi k R^3}{3} \end{aligned}$$

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4. A copper conductor has 8.5×10^{28} electrons per cubic meter available to carry current, i.e., the carrier density is $8.5 \times 10^{28} \text{m}^{-3}$. The drift velocity is found to be $3 \times 10^{-6} \text{m/s}$.

(a) Look up and report the charge of an electron in SI units.

(Partial) Solution/Hint →

$$|e| = 1.60217662 \times 10^{-19} \text{ Coul}$$

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- (b) [3 pts.] Calculate the current density J .

(Partial) Solution/Hint →

$$\begin{aligned} J = qnv &= 1.60217662 \times 10^{-19} \times 8.5 \times 10^{28} \times 3 \times 10^{-6} \\ &= \text{whatever} \end{aligned}$$

No unit conversions are necessary because the inputs are all in SI units. Whatever number you get by multiplying, will be in the correct SI units, which for the current density is A/m^2 .

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5. The current density in some region is found to be

$$\mathbf{J} = \frac{K_0 y}{\sqrt{x^2 + y^2}} \hat{i} - \frac{K_0 x}{\sqrt{x^2 + y^2}} \hat{j}$$

Use the continuity equation to show that the charge density in this region does not change with time.

(Partial) Solution/Hint $\rightarrow$

According to the continuity equation,

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0 \quad \implies \quad \frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}$$

Calculating the divergence of the given current density $\mathbf{J}$, one should obtain zero. That would show that the rate of change of charge density ($\partial \rho / \partial t$) is zero, i.e., that the charge density in this region does not change with time.

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